

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)	(i)		A {disease / it} that may be {passed / spread / transmitted / transferred} from {one individual to another / organism to organism / between individuals} Ignore any reference to mode of transmission	1			1		
		(ii)		Uses {weakened / killed / attenuated} pathogen / antigens / toxins} to stimulate {the immune response / production of antibodies / production of memory cells}	1			1		
		(iii)		(As transmitted through) {coughing and sneezing / droplet infection / airborne / aerosol} (1) overcrowded conditions / high population density areas / {high levels / increased risk} of contact (with infected people) (1)		1	1	2		
	(b)	(i)		A description of graph antibody levels constant / zero + then increase + then decrease (1) Explanation – any three (x1) from: B latent period / {macrophages / APC} carry out phagocytosis / antigen presentation takes place (1) C clonal selection / activation of B cells / T helpers release cytokines to stimulate B cells / B cells undergo clonal {expansion / proliferation} (1) D (increases as) plasma cells release antibodies (1) E (decreases as) antibodies are broken down (1)	2	2		4		